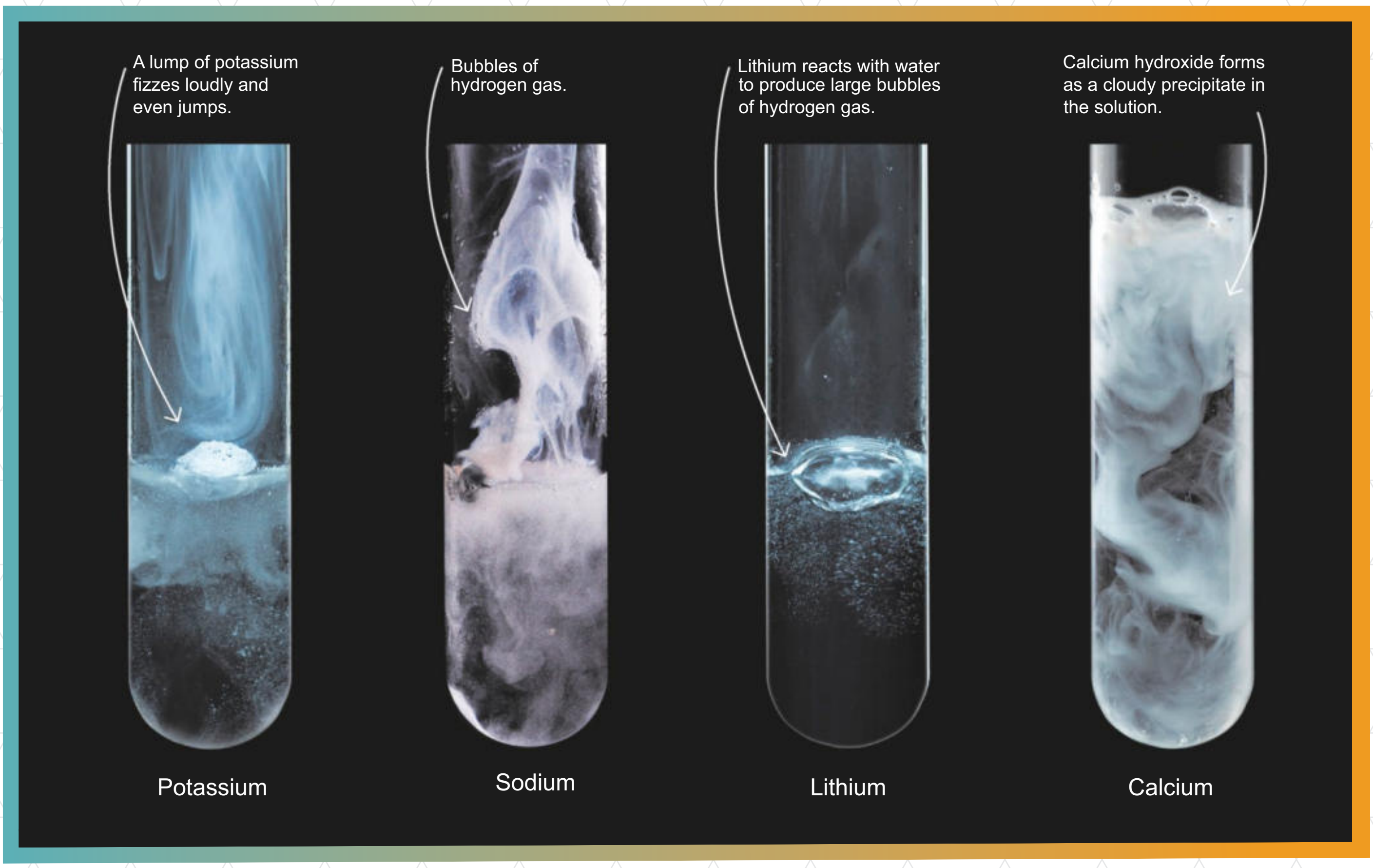
UNIT 08

CHEMICAL REACTIVITY



CHEMICAL REACTIVITY: the tendency of a substance to reacts chemically with another substance is called chemical reactivity.

METALS:

All those elements which possess the tendency to lose their valence electron and form cations or in other words generally metals are all those substances which are highly electropositive.

Properties of metals:

We already discussed their properties in chapter #04, Chemical bonding page no.49.

METALLIC CHARACTER OR ELECTROPOSITIVITY:

The tendency of an atom to lose their valence electron. It is the opposite to electro-negativity. Electropositivity increases in a group a and decrease in a period.

IONIZATIONPOTENTIAL/ENTHALPY/ENERGY:

The amount of energy which is required to remove an electron from a gaseous state or a specific atom. Alkali metals versus alkaline earth metals: Alkali metals have low Ionization Energy values than Alkaline earth metals, so Alkali metals are highly reactive than Alkaline earth metals.

SODIUM METAL:

Abundancy: 6th most abundant metal.
Constitution: 2.87% of earth's crust.
Position in periodic table:
Group IA.
Period 3rd.
Physical properties:
Na is silvery white.
Its melting point is 97.8 °C.
Its boiling point is 881.4 °C.

MAGNESIUM METAL:

Abundancy: 8th most abundant metal.
Constitution: 2.87% of earth's crust.
Position in periodic table:
Group IIA.
Period 3rd.
Physical properties:
Mg is grey-white metal.
Its melting point is 650 °C.
Its boiling point is 1090 °C.

CALCIUM METAL:

Abundancy: 5th most abundant metal.
Position in periodic table:

Group IIA.
Period 4th.
Physical properties:
Ca is white soft metal.
Its melting point is 851 °C.
Its boiling point is 1484 °C.

HARD METALS:
The metals which cannot be scratched easily are hard metals.
For example., Iron, Copper etc.

SOFT METAL:
The metal which can be scratched easily are soft metals.
For example., Sodium, Potassium etc.

NOBLE METALS:
Noble metals are all those metals which show inertness towards the other metals or non-metals because they don't oxidize easily.
Noble metals include Gold (Au), Silver (Ag), Platinum (Pt), Iridium (Ir), Osmium (Os), Rhodium (Rh), Ruthenium (Ru), Palladium (Pd).

NON-METALS:
All those elements which possess the tendency to gain the valence electron and form anions or in other words generally non-metals are all those substances which are highly electronegative.

PHYSICAL PROPERTIES OF NON-METALS:
i. Solids non-metals are brittle (break easily).
ii. Non-metals are bad conductors of heat and electricity (except graphite).
iii. They are not shiny, they are dull except iodine (it is lustrous like metals).
iv. They are generally soft (except diamond).
v. They have low melting and boiling points (except silicon, graphite and diamond).
vi. They have low densities.

CHEMICAL PROPERTIES OF NON-METALS:
i. Their valence shells are deficient of electrons, therefore, they readily accept electrons to complete their valence shells and become stable.
ii. They form ionic compounds with metals and covalent compounds by reacting with other non-metals e.g. CO_2, NO_2 atc.
iii. Non-metals usually do not react with water.
iv. They do not react with dilute acids because non-metals are itself electron acceptors.

COMMON CHEMICAL COMPOUNDS

(Common name, Chemical name, Chemical formula & Use):

Soda Ash

Na_2CO_3

Sodium Carbonate

Used as a water softener

Baking Soda

$NaHCO_3$

Sodium Bicarbonate

Used in baking powder, health salt, beverages

Table Salt

NaCl

Sodium Chloride

Used in food items

Sodium Nitrate

$NaNO_3$

Sodium Nitrate

Used as fertilizer and in dynamite

Slaked Lime

Used as soil conditioner, water treatment, steel industry

$Ca(OH)_2$

Calcium Hydroxide

Gypsum

$CaSO_4 \cdot 2H_2O$

Calcium Sulfate Dihydrate

Used in construction and plaster for setting broken bones

Calcium Hypochlorite

$Ca(OCl)_2$

Calcium Hypochlorite

Used in sterilizing water in swimming pools

Limestone

$CaCO_3$

Calcium Carbonate

Used in cement industry and as a source of CO_2